

STROKE NER ANNOTATION GUIDELINE

1. Project Overview

This guideline defines the annotation rules for a Named Entity Recognition (NER) dataset built from Indonesian stroke patient queries collected from Alodokter.com : one of Indonesia's largest online health consultation platforms. The dataset is the first stroke-specific NER corpus in the Indonesian language.

Parameter	Value
Source	Alodokter.com : online health consultation platform
Language	Indonesian (Bahasa Indonesia) : informal patient-generated text
Domain	Stroke and hemorrhagic stroke
Total cleaned dataset	1,742 patient queries
Tasks annotated	1,000 tasks (this guideline version)
Annotation tool	Label Studio
Tagging scheme	BIO : Beginning / Inside / Outside
Entity types	4 : SYM, DGN, TMP, RF

2. Entity Type Definitions

Four entity types are defined, all clinically relevant to stroke triage and early symptom recognition:

Tag	Full Name	Definition	Indonesian Example	English Translation
SYM	Symptoms	Any symptom, functional deficit, or physical manifestation reported by the patient or caregiver as a consequence of stroke.	<i>lemas, kesemutan, tidak bisa digerakkan, sulit bicara, mulut miring, mata melotot, afasia</i>	<i>weakness/fatigue, numbness/tingling, cannot be moved, difficulty speaking, mouth drooping, eyes bulging, aphasia</i>
DGN	Diagnosis	The stroke diagnosis or a clinical condition identified as a direct consequence of stroke. Includes lay descriptions of stroke pathology.	<i>stroke, stroke ringan, stroke iskemik, pembuluh darah pecah, pendarahan di otak, penyumbatan di otak</i>	<i>stroke, mild stroke, ischemic stroke, ruptured blood vessel, brain hemorrhage, brain blockage</i>
TMP	Temporal	Any time expression describing when symptoms started, how long they lasted, or the timing of stroke onset.	<i>tiba-tiba, 3 hari yang lalu, sejak kemarin, sudah 2 minggu</i>	<i>suddenly, 3 days ago, since yesterday, already 2 weeks</i>
RF	Risk Factors	Pre-existing medical conditions, patient age, lifestyle factors, or traumatic events that increase stroke risk.	<i>hipertensi, diabetes, usia 65 tahun, kolesterol, kecelakaan, perokok aktif</i>	<i>hypertension, diabetes, age 65 years, cholesterol, accident/trauma, active smoker</i>

3. BIO Tagging Scheme

The BIO (Beginning–Inside–Outside) scheme is used to annotate multi-token entity spans:

Tag	Meaning	Usage
B-[LABEL]	Beginning	The first token of an entity span.
I-[LABEL]	Inside	Any continuation token within the same entity span.
O	Outside	Any token that does not belong to any entity.

Example annotation:

Token (Indonesian)	BIO Tag	English Translation
<i>mama</i>	O	<i>my mother</i>
<i>saya</i>	O	<i>(possessive particle)</i>
<i>tiba-tiba</i>	B-TMP	<i>suddenly</i>
<i>tidak</i>	B-SYM	<i>cannot</i>
<i>bisa</i>	I-SYM	<i>(continuation)</i>
<i>bicara</i>	I-SYM	<i>speak</i>
<i>dan</i>	O	<i>and</i>
<i>tangannya</i>	O	<i>her hand</i>
<i>lemas</i>	B-SYM	<i>weak/limp</i>

4. Annotation Rules

4.1–4.6 Core Rules

Rule	Topic	Decision	Example (ID → EN)
4.1	Blood pressure values	Label the condition name (hipertensi, darah tinggi) as RF. Numeric values like 190/100 or GDS 204 are always O.	<i>hipertensi</i> → RF 190/100 → O <i>hypertension</i> → RF 190/100 → O
4.2	Patient age	Label the entire phrase usia/umur + number + tahun/th/thn as RF. Never split the phrase.	<i>usia 65 tahun</i> → RF umur 40th → RF <i>age 65 years</i> → RF age 40 → RF
4.3	Body part locations	Body parts alone (tangan, kaki, kepala) are always O. Exception: inseparable clinical expressions like mata melotot or mulut miring are labeled SYM.	<i>tangan</i> → O mata melotot → SYM <i>mulut miring</i> → SYM hand → O <i>eyes bulging</i> → SYM mouth drooping → SYM
4.4	Stroke spelling variants	All spelling variants of 'stroke' in Indonesian informal text are labeled B-DGN.	<i>struk, strok, storke, stoke, setruk, sroke</i> → DGN All variants = stroke diagnosis
4.5	Bangun tidur	The phrase 'bangun tidur' (woke up from sleep) is always O : it is an	<i>bangun tidur</i> → O woke up → O

		activity context, not a temporal marker for stroke onset.	
4.6	Sudah + duration	sudah + duration → TMP. sudah alone or sudah + recovery description → O.	sudah 3 hari → TMP sudah sembuh → O already 3 days → TMP already recovered → O

5.7–5.19 Extended Rules (added during annotation)

Rule	Topic	Decision	Example (ID → EN)
5.7	Negated symptoms	Functional deficits (ex : 'cannot speak') → SYM even when negated. Absent symptoms (ex : 'no vomiting') → O.	tidak bisa bicara → SYM tidak muntah → O cannot speak → SYM no vomiting → O
5.8	Lifestyle risk factors	Unhealthy lifestyle descriptions that increase stroke risk → RF.	makan gak terkontrol → RF jarang olahraga → RF uncontrolled eating → RF rarely exercises → RF
5.9	Slash in DGN	When two diagnoses are written with a slash, label only the first term as DGN. The slash and second term are O.	stroke ringan/parsial → only 'stroke ringan' = DGN mild/partial stroke → only first = DGN
5.10	Duplicate age spans	If patient age is mentioned more than once, keep only the most complete span (with usia/umur prefix). Remove duplicates.	usia 57 tahun + 57thn → keep 'usia 57 tahun', remove '57thn' keep most complete form
5.11	Recovery descriptions	Phrases describing recovery progress → O, not SYM.	tidak kunjung sembuh → O sudah membaik → O not recovering → O already improving → O
5.12	Negated risk factors	Risk factors preceded by a negation → O (condition is absent).	tidak ada hipertensi → O tidak punya DM → O no hypertension → O no diabetes → O
5.13	Medical procedures	Medical tests and procedures → O, not DGN.	CT scan → O MRI → O USG → O CT scan → O MRI → O
5.14	Numeric lab values	Numeric laboratory or measurement values → O. Label the condition name instead.	GDS 204 → O 178/110 → O hipertensi → RF blood glucose 204 → O hypertension → RF
5.15	Bell's Palsy	Bell's Palsy → DGN. It is a neurological diagnosis that co-occurs with stroke.	Bell's Palsy → DGN Bell's Palsy → DGN
5.16	Two symptoms joined by 'dan'	If two symptoms are connected by 'dan' (and), label each as a separate SYM span. 'dan' itself is O.	kram dan bergerak sendiri → two SYM spans cramps and moving by itself → two SYM spans
5.17	Menjalar	'Menjalar' (radiating/spreading) in context with a symptom → include in SYM span. Alone → O.	kebas menjalar ke jari → one SYM span menjalar alone → O numbness radiating to fingers → SYM
5.18	Tekanan darah / tensi	As an underlying condition → RF. Never SYM.	tekanan darah tinggi → RF tensi naik → RF high blood pressure → RF blood pressure rising → RF

5.19	Truncated text	If the text is cut off mid-sentence and the entity meaning is ambiguous → O. Do not guess.	<i>ambiguous token at end of truncated text</i> → O <i>ambiguous truncated token</i> → O
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5. Additional Edge Cases (decided during annotation)

These cases were decided during annotation sessions and are consistent with the rules above, but were not formally numbered. They should be treated as binding decisions.

Expression (Indonesian)	Label	Reasoning (English)
<i>2x, 3 kali, 5 kali seminggu</i>	O	Frequency expressions (2 times, 3 times/week) are not temporal markers for stroke onset → O
<i>masih bisa digerakkan</i>	O	Positive function (still can be moved) is NOT a symptom → O
<i>kepalanya terbentur (head trauma)</i>	RF	Head trauma is a risk factor for hemorrhagic stroke → RF
<i>cairan di otak</i>	DGN	Lay description of brain hemorrhage/hematoma → DGN
<i>stroke stadium 1</i>	DGN	Patient uses cancer staging terminology for stroke; label as-is → DGN
<i>tensinya turun (BP dropping)</i>	RF	Low blood pressure as a reported risk context → RF (edge case, note in paper)
<i>afasia</i>	SYM	Aphasia is a consequence/symptom of stroke, not a separate diagnosis in this context → SYM
<i>abses otak (brain abscess)</i>	SYM	Brain abscess caused by stroke = consequence of stroke → SYM
<i>edema serebri / edema cerebri</i>	SYM	Cerebral edema caused by stroke = consequence of stroke → SYM
<i>avc cerebral</i>	DGN	French abbreviation for AVC (accident vasculaire cérébral) = stroke → DGN
<i>encephalomalacia</i>	O	Can be caused by stroke, head trauma, or infection : context not specific to stroke → O
<i>penyumbatan darah di otak (57%)</i>	DGN	Already-diagnosed blockage reported with measurement = stroke diagnosis → DGN; numeric % → O

6. Indonesian–English Translation Reference

This section provides translations of the most frequent entity expressions in the dataset for non-Indonesian-speaking reviewers and collaborators.

6.1 SYM : Frequent Symptom Expressions

Indonesian	English	Clinical Note
<i>lemas</i>	weakness/fatigue/limp	Most frequent SYM (92x). General weakness of limbs.
<i>kesemutan</i>	tingling/pins and needles	36x. Paresthesia : common stroke symptom.

<i>pusing</i>	dizziness/vertigo/spinning	37x. Can indicate brainstem or cerebellar stroke.
<i>mati rasa</i>	numbness/loss of sensation	30x. Sensory deficit : classic stroke symptom.
<i>kejang</i>	seizure/convulsion	29x. Post-stroke seizure.
<i>kebas</i>	numbness/numb feeling	24x. Informal term for mati rasa.
<i>lumpuh</i>	paralysis/paralyzed	20x. Motor deficit : classic stroke symptom.
<i>kaku</i>	stiff/rigid/tense	18x. Muscle rigidity or spasticity.
<i>tidak bisa berjalan</i>	cannot walk	18x. Gait difficulty : motor deficit.
<i>tidak bisa digerakkan</i>	cannot be moved	14x. Severe motor deficit.
<i>sulit bicara</i>	difficulty speaking	15x. Speech impairment / dysarthria.
<i>sakit kepala</i>	headache	19x. Can indicate hemorrhagic stroke.
<i>pelo</i>	slurred speech	Informal term for dysarthria.
<i>mulut miring</i>	mouth drooping/asymmetric mouth	Facial droop : classic FAST stroke sign.
<i>mata melotot</i>	eyes bulging/eyes wide open	Fixed gaze : neurological symptom.
<i>afasia</i>	aphasia	Language disorder caused by stroke.
<i>tidak sadarkan diri</i>	loss of consciousness/unresponsive	Severe stroke symptom.
<i>drodog</i>	tremor/shaking (Javanese)	Javanese lay term for tremor.
<i>klenger</i>	fainting/collapse (Javanese)	Javanese lay term for sudden collapse.
<i>bengo</i>	confused/disoriented (Javanese)	Javanese lay term for confusion.
<i>pelo / cadel / celat</i>	slurred speech/speech impairment	Multiple informal surface forms for dysarthria.
<i>turun bero</i>	sudden drop/drastic deterioration	Regional informal term for sudden worsening.
<i>urat2 mengeras</i>	muscles hardening (Javanese/informal)	Informal description of muscle spasticity.

6.2 DGN : Frequent Diagnosis Expressions

Indonesian	English	Clinical Note
<i>stroke</i>	stroke	875x. Standard term.
<i>stroke ringan</i>	mild stroke/TIA	183x. Most common subtype label.
<i>stroke iskemik</i>	ischemic stroke	21x. Clinical subtype.

<i>stroke hemoragik / hemorrhagic</i>	hemorrhagic stroke	Multiple spellings.
<i>struk, strok, storke, stoke, setruk</i>	stroke (misspellings)	20+ informal spelling variants of 'stroke'.
<i>pembuluh darah pecah</i>	ruptured blood vessel	Lay description of hemorrhagic stroke.
<i>penyumbatan di otak</i>	brain blockage/occlusion	Lay description of ischemic stroke.
<i>pendarahan di otak</i>	brain hemorrhage/bleeding in the brain	Lay description of hemorrhagic stroke.
<i>penyumbatan pada batang otak</i>	brainstem blockage	Brainstem stroke description.
<i>cairan di otak</i>	fluid in the brain	Lay description of hematoma/hemorrhage.
<i>Bell's Palsy</i>	Bell's Palsy	Facial nerve palsy, co-occurs with stroke.
<i>stroke batang otak</i>	brainstem stroke	Specific anatomical subtype.
<i>avc cerebral</i>	AVC (French: accident vasculaire cérébral)	French abbreviation for stroke : rare finding.

6.3 RF : Frequent Risk Factor Expressions

Indonesian	English	Clinical Note
<i>hipertensi</i>	hypertension/high blood pressure	55x. Most frequent RF.
<i>darah tinggi</i>	high blood (informal for hypertension)	32x. Most common lay term for hypertension.
<i>tekanan darah tinggi</i>	high blood pressure	13x. More formal variant.
<i>diabetes</i>	diabetes	30x. Type 1 or 2, not distinguished.
<i>diabetes melitus</i>	diabetes mellitus	Formal clinical term.
<i>kolesterol</i>	cholesterol (high cholesterol)	19x. Hypercholesterolemia.
<i>jantung / penyakit jantung</i>	heart disease/cardiac condition	14x. Cardiovascular RF.
<i>pengentalan darah</i>	blood clotting/hypercoagulability	6x. Coagulation disorder.
<i>kecelakaan</i>	accident/trauma	7x. Head trauma as hemorrhagic stroke RF.
<i>asam urat</i>	gout/uric acid	4x. Metabolic comorbidity.
<i>perokok / perokok aktif</i>	smoker/active smoker	Lifestyle RF.
<i>hipotensi</i>	hypotension/low blood pressure	Edge case RF : low BP can trigger stroke.
<i>trombositopenia</i>	thrombocytopenia	Low platelet count as RF.
<i>usia [N] tahun / umur [N] th</i>	age [N] years old	Patient or family member age as RF.

6.4 TMP : Frequent Temporal Expressions

Indonesian	English	Clinical Note
<i>tiba-tiba</i>	suddenly/all of a sudden	52x. Most frequent TMP : indicates acute onset.
<i>tiba2/tiba tiba</i>	suddenly (informal)	27x/18x. Informal variants of tiba-tiba.
<i>mendadak</i>	suddenly/abruptly	13x. Formal synonym for tiba-tiba.
<i>3 hari yang lalu</i>	3 days ago	Duration since onset.
<i>sudah 2 minggu</i>	already 2 weeks	Duration with 'sudah' (already).
<i>kemarin</i>	yesterday	Recent onset marker.
<i>tadi pagi/pagi tadi</i>	this morning/earlier this morning	Time of day onset.
<i>sejak [time]</i>	since [time]	Duration from a reference point.
<i>beberapa hari ini</i>	these past few days/lately	Vague recent duration.
<i>baru saja</i>	just now/just recently	Very recent onset.
<i>habis sahur</i>	after sahur (pre-dawn meal during Ramadan)	Islamic calendar time reference.

7. Dataset Statistics : 1,000 Tasks

Label	Count	%	Unique	Hapax (%)	Top Expression
SYM : Symptoms	1662	39.5%	861	79%	<i>lemas (92x)</i>
DGN : Diagnosis	1381	32.8%	162	70%	<i>stroke (875x)</i>
TMP : Temporal	645	15.3%	404	82%	<i>tiba-tiba (52x)</i>
RF : Risk Factors	522	12.4%	303	84%	<i>hipertensi (55x)</i>
TOTAL	4,210	100%	1,730	:	

8. Zero-Entity Policy

Some tasks in the dataset contain no labeled entities. These are intentional and valid. A task should have zero entities if:

Condition	Example (English)
The query is about a non-stroke condition (TB, pregnancy, liver disease, etc.)	"My father has tuberculosis and is coughing a lot" → <i>O</i>
The query is a purely mechanistic or educational question with no symptom report	"What causes Broca's aphasia?" → <i>O</i>

The query asks about medication dosages or procedures without reporting symptoms	<i>"Can a stroke patient take vitamin D?" → O</i>
The clinical term could be caused by multiple conditions, not specifically stroke	<i>"What is Encephalomalacia?" → O (could be stroke, trauma, or infection)</i>
The text is too truncated to identify any entity with confidence	<i>Text cut off mid-sentence with ambiguous context → O</i>

Zero-entity rate in this dataset: 39/1,000 tasks (3.9%). All zero-entity tasks were verified manually.

9. Quality Control Summary

Batch	Tasks	Clean Rate	Key Issues Fixed
Tasks 1–100	100	100%	Whitespace, slash DGN, duplicate age, recovery descriptions
Tasks 101–200	100	99%	CT scan → O (×5), numeric BP → O (×4), age boundary (×1)
Tasks 201–300	100	97%	CT scan as DGN, duplicate age (×2)
Tasks 301–400	100	97%	Numeric BP as RF, duplicate age, whitespace TMP
Tasks 401–503	103	99%	Duplicate age, stroke as SYM → DGN
Tasks 504–600	97	99%	Duplicate age + vertigo RF span
Tasks 601–700	100	100%	Zero errors : first perfect batch
Tasks 701–803	103	99%	Duplicate age (Inner 798), numeric BP (Inner 841)
Tasks 804–900	97	97%	Numeric BP → O, duplicate pendarahan intra cerebrum
Tasks 901–1000	100	98%	Duplicate age (Inner 957, 959)
CUMULATIVE	1000	~98.5%	39 zero-entity tasks (3.9%) : all verified